

21ECP401L-MAJOR PROJECT

ATTENTION-ENHANCED U-NET WITH TRANSFORMER ENCODER FOR FETAL HEAD SEGMENTATION AND GROWTH PARAMETER ESTIMATION

A PROJECT REPORT

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**In partial fulfillment of the requirements for the Degree
of**

BACHELOR OF TECHNOLOGY

In

**ELECTRONICS & COMMUNICATION ENGINEERING
WITH SPECIALIZATION IN CYBER PHYSICAL SYSTEMS**



**DEPARTMENT OF ELECTRONICS AND
COMMUNICATION ENGINEERING**

COLLEGE OF ENGINEERING AND TECHNOLOGY

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

KATTANKULATHUR-603 203

MAY 2026

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

(Under Section 3 of UGC Act, 1956)

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

BONAFIDE CERTIFICATE

This is to certify that the project report titled “**ATTENTION-ENHANCED U-NET WITH TRANSFORMER ENCODER FOR FETAL HEAD SEGMENTATION AND GROWTH PARAMETER ESTIMATION**”, submitted for **21ECP401L – Major Project**, is a Bonafide work carried out by the following students:

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DECLARATION

We hereby declare that the Major Project entitled “**ATTENTION-ENHANCED U-NET WITH TRANSFORMER ENCODER FOR FETAL HEAD SEGMENTATION AND GROWTH PARAMETER ESTIMATION**” to be submitted for the Degree of Bachelor of Technology is our original work as a team and the dissertation has not formed the basis of any degree, diploma, associateship or fellowship of similar other titles. It has not been submitted to any other University or institution for the award of any degree or diploma.

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ACKNOWLEDGEMENTS

We would like to express our deepest gratitude to the entire management of SRM Institute of Science and Technology for providing us with the necessary facilities for the completion of this project.

We wish to express our deep sense of gratitude and sincere thanks to our Professor and Head of the Department, **Dr. M. Sangeetha**, for her encouragement, timely help, and advice offered to us.

We are very grateful to our supervisor **Dr. Manikandan A**, Assistant Professor, Department of Electronics and Communication Engineering, for providing invaluable guidance with inspiring dedication, untiring efforts, and tremendous enthusiasm in making this project successful and presentable.

We would like to express our sincere thanks to the project coordinator **Dr. Anilet Bala A** and panel members **Dr. Latha S** and **Dr. Sayantani Bhattacharya** for their time and suggestions for the implementation of this project.

We also extend our gratitude and heartfelt thanks to all the teaching and non-teaching staff of the Electronics and Communications Engineering Department and to our parents and friends, who extended their kind cooperation using valuable suggestions and timely help during this project work.

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ABSTRACT

Gestation is a significant process in the prenatal care, to track the progress of a fetus. The study of fetal head ultrasound examination is a routine activity of physicians to determine such parameters as Head Circumference (HC), Biparietal Diameter (BPD), Occipitofrontal diameter (OFD). Estimates of the age of a fetus during gestation and diagnosis of developmental abnormalities (microcephaly and macrocephaly) are made using these readings. The measurements are currently done manually by qualified individuals in the clinical practice. Nevertheless, manual analysis of ultrasound images is not only tedious but also based on subjective results despite the fact that it can yield good outcomes, it relies on the experience of an operator.

To solve this problem, we introduce an automated system comprising of a convolution neural network (CNN) based architecture of fetal head detection and biometric measure of ultrasound images. The model in this paper employs a convolutional neural network in the form of an Attention U-Net to detect and localize the fetal skull region in the axial ultrasound images. Head boundary is then obtained and geometry analysis is further done upon receiving segmentation. Next, there are head circumference, biparietal diameter, occipitofrontal diameter and area that are clinically relevant parameters extracted by techniques such as contour detection and the ellipse fitting.

Sustainable Development Goal



The proposed system supports SDG 3 by improving early detection of abnormal fetal development through accurate automated ultrasound analysis while reducing manual errors. It contributes to SDG 9 by enhancing diagnostic accuracy and promoting intelligent, scalable digital healthcare infrastructure using AI-driven solutions. It also aligns with SDG 10 by enabling reliable fetal growth monitoring in rural and resource-constrained healthcare settings with reduced dependence on highly skilled radiologists.

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LIST OF SYMBOLS AND ABBREVIATIONS

Symbol / Abbreviation	Description
a	Semi-major axis of the fitted ellipse (half of the OFD)
b	Semi-minor axis of the fitted ellipse (half of the BPD)
BCE	Binary Cross-Entropy (Loss function for pixel-wise classification)
BPD	Biparietal Diameter (The width of the fetal head)
CNN	Convolutional Neural Network
DICOM	Digital Imaging and Communications in Medicine
FHR	Fetal Head Region
GA	Gestational Age (Estimated age of the fetus)
HC	Head Circumference (The perimeter of the fetal head)
MAE	Mean Absolute Error (Metric for measurement accuracy)
Npixels	Number of pixels classified as the anatomy within the mask
OFD	Occipitofrontal Diameter (The longitudinal length of the head)
π	Pi (Constant used in circumference and area calculations)
pixel_size	Spatial resolution (Physical size of one pixel in mm)
ReLU	Rectified Linear Unit (Activation function in the U-Net/ResNet)
ResNet50	Residual Network (50-layer deep learning architecture)
U-Net	U-shaped Network (Architecture for biomedical image segmentation)
$\approx$	Approximately equal to (Used in Ramanujan's HC estimation)

CHAPTER 1

INTRODUCTION

1.1 GENERAL

Gestational age can also be determined through a technique known as fetal biometry, which can be valuable for assessing fetal growth and early developmental disorders. The main biometric indexes for diagnosing gestation age and detecting genetic disorders, such as microcephaly and macrocephaly are head circumference (HC), biparietal diameter (BPD) and occipitofrontal diameter (OFD). Given that fetal head axial ultrasound examinations are the basis of most prenatal screening systems, these parameters are typically obtained as part of such procedures.

Ultrasound imaging is the most common, most widely used and non-invasive modality for fetal evaluation. However, the accurate measurement of fetal head parameters depends on the skills of sonographers and clinicians. Intra- and interobserver variability has been demonstrated for the manual measurement of skull boundary with caliper, which can be a very painstaking and biased process. Specially, speckle noise, low contrast, acoustic shadowing and fetal motion add more difficulties to the accurate measurement. Such flaws reveal the need for fast and reliable computer-aided diagnostic systems.

1.2 CHALLENGES IN ULTRASOUND IMAGE ANALYSIS

While widely used, ultrasound imaging is challenging for computer-aided analysis. The nature of ultrasound images is subject to speckle, poor contrast, shadowing and blurriness. This can hinder the visualisation of anatomical structures, particularly during early pregnancy.

Additionally, the position, orientation and movements of the fetus complicate the process of locating and segmenting the region of interest. The fetal head, while roughly elliptical in shape, can be distorted by imaging factors and is difficult to delineate. These factors require the design of efficient and smart algorithms that are able to handle these variations without compromising accuracy.

1.3 NEED FOR AUTOMATED FETAL BIOMETRY SYSTEMS

Fetal biometry techniques can be used to assess the growth of the fetus, estimate gestational age, and detect abnormalities like microcephaly and macrocephaly. Typically, clinicians will locate specific landmarks on the body and measure with calipers using the periphery of the fetus' head. However, such a process can be very time-consuming and highly operator-dependent.

To overcome such challenges, automated fetal biometry tools were introduced into clinical practice. These tools are algorithmic and therefore minimize human error. Automated tools are important in high throughput clinics since they ensure accuracy and fast measurements. Such tools can assist novices to get reliable measurements.

1.4 ROLE OF DEEP LEARNING IN MEDICAL IMAGE SEGMENTATION

The current trend for medical image analysis is to adopt deep learning, a type of machine-learning algorithm that learns high-level features from the data it receives instead of hand-crafted features. Due to the invariance of their parameters with respect to translation, convolutional neural networks (CNNs) have widely been used in segmentation tasks- that is assigning a class label for each pixel of an image[1]. Tailored U-Net architectures for nuanced biomedical image segmentation have shown great performance on medical imaging tasks (live-cell microscopy, histopathology super-pixel segmentation).

Instead of complex networks, U-Net architecture is a simple encoder-decoder network with skip connections to keep some information spatial. The encoder learns features based on context, and the decoder reconstructs your segmentation map. This type of design enables accurate localization within structures and in low-contrast images with noise. Due to the automatic feature-learning ability of deep learning models, they are fairly advantageous over traditional image analysis algorithms and have overcome many limitations (including speckle noise, weak boundaries, structural variations) which gulf fetal ultrasound images making them useful for automated fetal head segmentation.

1.5 MOTIVATION OF THE PROPOSED WORK

Although today's state-of-the-art imaging technologies such as ultrasound, and CT surface rendering-based methodologies offer relatively reliable measurements of the fetal head size, it remains challenging because of the interference in the measurement procedure due to noise and variability in the position of the fetus. Manual and semi-automatic segmentation

procedures commonly face difficulties in providing reliable results under such circumstances. Therefore, there is a need for an accurate system capable of accurately segmenting the fetal head and computing biometrics without much manual intervention. This research project aims to establish an integrated framework for deep learning-based segmentation combined with geometric computation. Through the combination of U-Net-based segmentation and ellipse-fitting techniques, this system is intended to provide reliable estimates of head circumference and its parameters.

1.6 OBJECTIVES OF THE STUDY

The main objectives of this study are as follows:

- To design and implement a deep learning model for accurate fetal head segmentation from ultrasound images.
- To extract key biometric parameters including head circumference (HC), biparietal diameter (BPD), and occipitofrontal diameter (OFD).
- To convert pixel-based measurements into real-world units using pixel spacing information.
- To evaluate the performance of the system using error metrics by comparing predicted values with reference measurements.
- To develop a visualization framework that displays measured parameters on segmented outputs.

1.7 SCOPE OF THE WORK

The purpose of this work was segmentation and quantification of fetal head structures in two-dimensional ultrasound images. Overview of the Study This project had four stages Preprocessing Model Training Segmentation Parameter Estimation HC, BPD, OFD and Head area measurement are some of the fundamental measurements in prenatal sonography [18].

It can still be referred to as a research model and not be considered a tool for actual clinical practices. It is also important to state that 3D ultrasound images as well as fetus anatomy is out of scope of this project.

CHAPTER 2

LITERATURE SURVEY

Fetal ultrasound imaging plays a crucial role in prenatal diagnosis, particularly for assessing fetal growth and detecting abnormalities. Accurate measurement of fetal head parameters such as head circumference (HC), biparietal diameter (BPD), and occipitofrontal diameter (OFD) is essential for estimating gestational age and identifying developmental issues. However, manual measurement is time-consuming, operator-dependent, and prone to variability. With the advancement of artificial intelligence and computer vision, deep learning techniques have been increasingly applied to automate fetal ultrasound analysis. Over the past decade, various approaches including convolutional neural networks (CNNs), attention-based models, and hybrid architectures have significantly improved segmentation accuracy and biometric estimation reliability.

Deep learning models, especially CNN-based architectures, have demonstrated strong capability in medical image analysis due to their hierarchical feature extraction ability. Qin et al. [1] proposed a CNN-based method for automatic fetal head circumference measurement, showing improved accuracy over traditional techniques. Similarly, Chen et al. [2] introduced an anatomy-aware attention network that enhances segmentation performance by focusing on relevant fetal head regions. Zhou et al. [3] utilized multi-task learning to simultaneously estimate multiple fetal biometric parameters, improving efficiency and consistency. Huang et al. [4] implemented a residual U-Net with attention mechanisms, demonstrating better boundary detection in ultrasound images. Wang et al. [5] and Namburete et al. [6] further validated the effectiveness of deep learning in automated fetal biometry, highlighting improved precision and reduced manual effort.

To enhance segmentation robustness, attention mechanisms and feature fusion strategies have been widely explored. Wu et al. [7] proposed an attention-based U-Net to improve segmentation performance under noisy conditions. Bano et al. [8] developed a deep learning framework for automated fetal head analysis, integrating segmentation and measurement tasks. Kumar et al. [9] presented a complete automated system for fetal biometry using deep learning techniques. Zhang et al. [10] introduced multi-scale feature fusion to improve measurement accuracy across varying image resolutions. These approaches demonstrate that combining attention mechanisms with deep architectures

significantly improves segmentation quality and robustness.

Recent research has focused on integrating transformer-based models into medical imaging to capture global contextual information. Li et al. [11] and Ahmed et al. [12] explored transformer-based segmentation methods, showing improved performance in handling complex ultrasound images. Park et al. [13] developed deep attention networks for robust head circumference measurement, while Zhao et al. [14] proposed automated skull contour extraction techniques for improved boundary detection. Singh et al. [15] demonstrated the effectiveness of ellipse fitting on segmentation outputs for accurate biometric estimation. These studies highlight the importance of combining segmentation with geometric modeling for precise parameter extraction.

More recent advancements emphasize hybrid and transformer-based architectures for improved performance. Ibrahim et al. [16] proposed a segmentation-driven framework combining deep networks for enhanced measurement accuracy. Patel et al. [17] developed a deep learning system for precise fetal biometric estimation using optimized architectures. Chen et al. [18] and Nguyen et al. [19] introduced attention-guided transformer networks, which significantly improve segmentation accuracy by capturing long-range dependencies in ultrasound images. Lin et al. [20] proposed real-time fetal head boundary detection systems using spatial attention mechanisms, demonstrating the feasibility of deploying such models in clinical environments.

Despite significant progress, several challenges remain in fetal ultrasound analysis. Many models rely heavily on image-based features without incorporating clinical constraints, which may affect robustness. Class imbalance and variability in ultrasound image quality pose additional difficulties for model generalization. Furthermore, transformer-based and hybrid models often require high computational resources, limiting their real-time applicability. Therefore, there is a need for efficient and accurate models that combine deep learning, attention mechanisms, and geometric analysis to achieve reliable segmentation and measurement.

The proposed work addresses the limitations of existing approaches by integrating a transformer-enhanced U-Net model for precise fetal head segmentation with geometric parameter extraction techniques. Unlike conventional methods that rely solely on convolutional features, the incorporation of transformer-based attention enables the model to capture both local spatial details and global contextual information, which is

essential for handling low-contrast and noisy ultrasound images. The segmentation output is further utilized to compute key fetal biometric parameters such as head circumference (HC), biparietal diameter (BPD), occipitofrontal diameter (OFD), and head area using ellipse fitting methods, ensuring accurate and consistent measurements. Additionally, gestational age estimation is derived from the computed head circumference, providing clinically useful insights for fetal growth assessment. To further enhance the system, a lightweight classification module is incorporated to indicate potential abnormal cases, supporting early-stage clinical decision-making without significantly increasing computational complexity. The overall framework forms a complete automated pipeline that reduces operator dependency, improves measurement accuracy, and ensures robustness across varying ultrasound image conditions. By combining deep learning, attention mechanisms, and geometric analysis, the proposed system offers an efficient, scalable, and clinically relevant solution for fetal ultrasound analysis and prenatal diagnostics.

CHAPTER 3

SYSTEM MODEL DESCRIPTION

3.1 INTRODUCTION

The system obtains fetal ultrasound images to allow automatic analysis of the images via computer software. Preprocessing of ultrasound images is done to enhance their quality through resizing, normalization, and noise reduction. The ultrasound images undergo encoding in both the spatial and texture features via an encoder network. By using the transformer encoder with the attention module, the machine can be able to incorporate global context and capture long-range dependencies like rewarding parts of the image containing the majority of the fetus head region. Decoding of features is achieved by the decoder network to form the proper segmentation of the fetus head mask. Post-processing techniques are used to remove any noise in the segmentation process to ensure clear borders of the image segments. After that, head contour detection is carried out to obtain the fetus head outline and extract the parameters from it. The rest parameters are HC, BPD, OFD, and Head Area. Figure 3.1 represents the system model block diagram.



Figure 3.1: System Model Description

3.2 DATA PREPROCESSING

The input data consists of the raw images of ultrasound (resolution = 800 x 540) and expert generated masks for the semantic labels of the fetus head. As part of preprocessing, which will help in efficient computation and consistency, the ultrasound images are converted to square images (512x512) matrix form. Moreover, normalization is applied to these images, resulting in improved convergence of the training process as well as making the model invariant to contrast variation in ultrasound images. Data partitioning refers to the division of preprocessed data into three categories viz Training, Validation, and Testing. The objective of this process is that the model should be able to train itself using one set, validate itself on another set and finally test itself on a completely new set.

3.3 DEEP LEARNING MODEL

The model makes use of the general architecture of U-Net for image segmentation, which has proved to be a benchmark in biomedical imaging. The model has a contracting path (Encoder) that captures visual information and creates feature maps through a bottleneck architecture, connected with an expanding path (Decoder). This helps the model localize the segmented head accurately. Training Approach: The model is trained on optimizing its weight through a hybrid loss function, i.e., Binary Cross Entropy (BCE) and Dice Loss function. BCE helps in evaluating the pixel wise accuracy while Dice Loss ensures overlapping between the segmented masks and ground truth label (maximize clarity of the body part) along with balancing the class imbalance issue between the background and the anatomy of interest. Dice coefficient for validation purpose. Segmentation Prediction Once trained, the model predicts a new mask for the image and gives raw binary masks that belong to the 'pass'(1)/ 'not-pass'(0) class 'head' pixel.

3.4 MORPHOLOGICAL PROCESSING & GEOMETRIC EXTRACTION

The outputs of neural networks may sometimes have some minor errors; therefore, mathematical morphology and geometric processing is adopted to obtain the correct data. Mask Post-processing: The mask is cleaned using a certain threshold level, thereby removing all those regions in the mask that have lower levels of confidence. An algorithm is employed to fill the small holes in the mask so as to obtain an entirely filled region representing the head of the fetus. Contour Extraction & Ellipse Fitting: The outer contour of the refined mask is detected, which is then used to represent the contour of the head. Since the head of the fetus has an elliptical shape, ellipse fitting is applied to the head contours. This converts the biological structure into geometry and establishes axes. Parameter Extraction & Pixel-to-mm Conversion: In the next step, the digital parameters of the fetus such as HC, BPD, OFD, and Head Area are calculated

based on the ellipsoidal model. To obtain these parameters in physical dimensions such as mm, a metadata file is required.

3.5 CLINICAL INFERENCE

These physical characteristics are translated to clinical findings.

Gestation age estimation: Lastly, the measured physical values (HC, BPD, and OFD) are compared to standards found in growth charts or equations for estimating GA.

Anomaly classification: The process carried out after identifying anomalies, the step is carried out in order to categorize the abnormalities in measurement patterns according to time as well as entry to the ensemble median during further follow-up. Statistics limits: Anything exceeding the limits considered acceptable in statistics is termed an abnormality.

3.6 ASSESSMENT AND REPORT

The software makes use of statistical analysis in comparing the predicted HC value against the clinically measured values for accuracy in the CSV file of the dataset. The software also measures its accuracy through errors such as the Mean Absolute Error (MAE) and percentage error. The pipeline creates a clinical package that includes a segmentation mask visualization, geometric measurements, prediction of gestational age, clinical diagnosis result (Normal/Abnormal), and results table.

The system's performance depends on ultrasound image quality, accurate annotations, and correct pixel spacing, and variations in equipment, noise, and operator technique can affect segmentation and measurement accuracy. The model may not generalize well to all populations without diverse training data. Patient data must be anonymized and handled with strict confidentiality, and the system is intended only for decision support, not as a replacement for clinical judgment, with all results requiring validation by medical professionals.

The implementation follows standard machine learning and medical imaging practices, including structured preprocessing, validation-based model selection, and evaluation using appropriate metrics. Reproducibility is maintained through consistent configurations and documented workflows, and the system is developed using established frameworks to ensure reliability, robustness, and consistency, aligning with general engineering and biomedical application standards.

CHAPTER 4

METHODOLOGY

4.1 DATASET

In order to analyze the suggested fetal head biometry approach, we use the set of transabdominal sonograms obtained during regular prenatal examinations, i.e., our curated set of fetal ultrasound images. The data contains grayscale ultrasound images and corresponding fetal head contours providing an adequate baseline for training segmentation. All images are presented in their native resolution of 800×540 pixels. The images were uniformly resampled to 512×512 pixels in order to simplify further work with them. In addition, pixel spacing in mm per pixel and the actual Head Circumference (HC) measurements taken from a metadata CSV file allow switching between millimeters and pixels-based scales. Table 4.1 shows us the various parameters and its description on how it is used.

Table 4.1: Dataset Description

Parameter	Description
Imaging Modality	2D Obstetric Ultrasound
Original Resolution	800×540 pixels
Training Resolution	512×512 pixels
Annotation Type	Binary Head Mask
Mask Format	Grayscale
Metadata Available	Pixel Size (mm), Clinical HC (mm)

4.2 PREPROCESSING

The goal of preprocessing is to achieve more accurate segmentation and ensure consistency across space. Firstly, the aspect ratio of all ultrasound images is preserved by padding and rescaling to 512×512 pixels, which makes it possible to provide inputs for the network accordingly. Optional intensity normalization will make it easier to reach more consistent results while training the model. The binary masks are also resized by means of applying the nearest neighbor interpolation in order to maintain high quality of boundaries. During the training process, several techniques of data augmentation such as horizontal flips, rotations, elastic deformations, and geometric distortions are implemented in order to boost the generalization capability of the network and prevent overfitting.

4.3 SEGMENTATION NETWORK ARCHITECTURE

We use a U-Net architecture based on convolutional neural network (CNN) to extract the fetal head region (FHR) of the obtained images. The design is based on an encoder-decoder architectural format with skip connections to maintain the fine spatial resolution. The encoder gradually removes features at several levels of abstraction, Every deep learning layer includes convolutional functions with subsequent batch-normalization and Rectified Linear Unit (ReLU) functions, and dropout is used to counter overfitting. In training, a hybrid Binary Cross-Entropy Dice loss is used to minimize the overlaps of regions and boundary delineation. The Adam optimizer is used to carry out training with dynamically modified learning rate, thus ensuring a stable convergence. Figure 4.1 displays the architecture of CNN model.

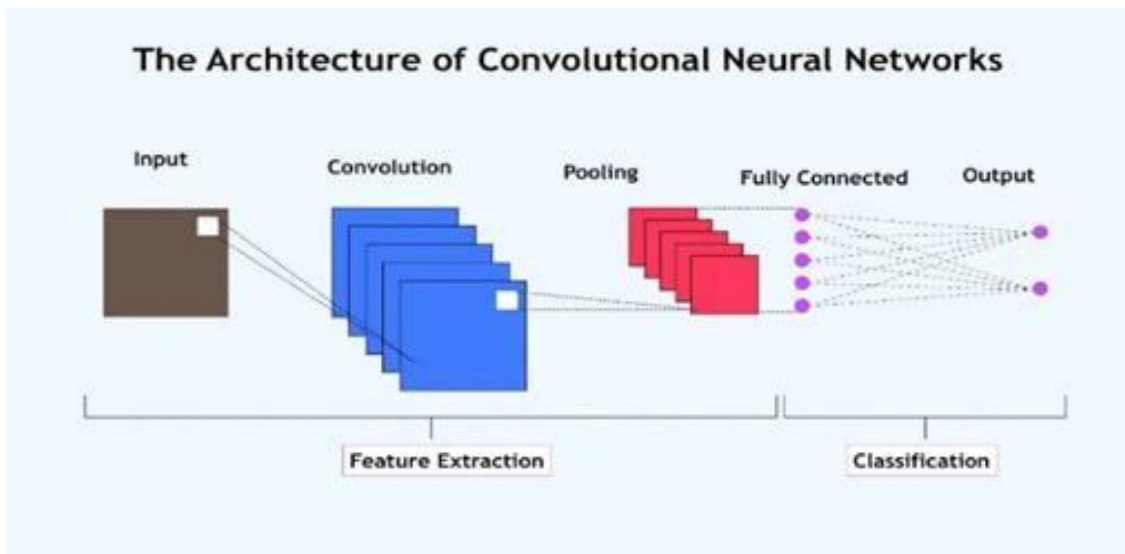


Figure 4.1: CNN Architecture

4.4 POST-PROCESSING AND GEOMETRIC ANALYSIS

After morphological operations, the predicted mask is then narrowed down to remove small artefacts and ensure a continuous region of the head. The most prominent connected part is isolated to isolate the fetal cranial structure. The results of the previous step are then followed by subsequent contour extraction using the boundary detecting algorithms and the head boundary is then approximated by the ellipses fitting. Since the elliptical geometry of the fetal head is estimated as elliptical in the existing axial ultrasound imaging, ellipse fitting, which is a geometrical parameter, can be easily estimated using the ellipse.

The large and small axes of the fitted ellipse are associated with the occipitofrontal and biparietal diameter respectively (OFD and BPD). The estimation of head circumference (HC) is done by applying the approximation of the perimeter of an ellipse by Ramanujan. The area is calculated as the pixel-based area of the extracted contour to obtain the head area.

Since the processed images are resized to 512 x 512, scaling correction is used to scale the measurements to the original picture dimension of 800 x 540. After this change pixel-based measurements are changed into millimetric measurements with the help of spatial resolution metadata of the image file.

4.5 BIOMETRIC PARAMETER ESTIMATION

The system automatically computes the following fetal biometric parameters:

Head Circumference (HC) – Estimated using ellipse perimeter approximation.

Biparietal Diameter (BPD) – Minor axis of fitted ellipse.

Occipitofrontal Diameter (OFD) – Major axis of fitted ellipse.

Head Area – Pixel area converted into mm².

To validate measurement accuracy, predicted HC values are compared with clinically record HC values are from the dataset. Absolute error and a percentage error metrics are calculated to quantified performance. Figure 4.2 displays the various parameters measured.



Figure 4.2: Fetal Head Parameters

4.6 ANATOMICAL METRIC USING GEOMETRIC FORMULATION

The below equations define the mapping of the segmented digital image domain into the standard biometric domain, specifically the head circumference of the fetus (other elliptical features too).

4.6.1 Head Circumference (HC)

Definition: The calculation of the Head Circumference through the determination of the ellipse's perimeter.

Mathematical Background: To find out the perimeter of an ellipse, we use the elliptic integral, which doesn't have an analytical solution and requires extensive time to compute. The present formula is based on the first approximation of the perimeter found by Ramanujan. It is very accurate when approximating the ellipse's perimeter with the use of semi-axes (a, b).

4.6.2 Biparietal Diameter (BPD) and Occipitofrontal Diameter (OFD)

$$BPD = 2b$$

$$OFD = 2a$$

Description: These equations represent the anatomical structure's major axes.

Context:

BPD (Biparietal Diameter) is the width of the head. This is the minor axis of the ellipse and is equal to twice the semi-minor axis (b).

OFD (Occipitofrontal Diameter) is the longitudinal length of the head, from the occipital bone to the frontal bone. This is equivalent to the major axis of the ellipse (twice the semi-major axis).

Digital to Physical Space Conversion: To make the extracted features clinically meaningful, measurements need to be translated from pixels to physical units (e.g., millimeters).

4.6.3 Area Calculation

Description: Calculate the total area of the segmented region.

Context: pixels is the absolute number of pixels that have been classified as the anatomy of interest in the segmentation mask. The 'pixel_size' (usually extracted from the DICOM metadata of the scan) is the physical size of a single pixel. The square of the pixel size gives the physical area of a pixel. The absolute area can be obtained by multiplying this by the number of pixels in the region.

$$HC \approx [3(a + b) - \sqrt{\{(3a + b)(a + 3b)\}}] \quad (4.1)$$

$$BPD = 2b \quad (4.2)$$

$$OFD = 2a \quad (4.3)$$

$$Head\ Area = N_{\{pixels\}} \times (pixel_size)^2 \quad (4.4)$$

$$Measurement_{\{mm\}} = Measurement_{\{pixels\}} \times pixel_size \quad (4.5)$$

where,

a = semi-major axis

b = semi-minor axis

N_pixels = segmented area in pixels

CHAPTER 5

U-NET ARCHITECTURE

U-Net is a convolutional neural network that is utilized in the segmentation of biomedical images. It is designed architecturally as encoder-decoder system, in which the encoder phase generates features of the input image by performing successive down sampling procedures, hence retrieving salient global information. The decoder step then applies up sampling to the reconstruction process to release spatial information, after which, the creation of the segmentation output is possible. Another characteristic feature of U-Nets is skipping connections between the similar layers between the encoder and decoder. These skip connections maintain high spatial resolution dimensions hence relaying finer details between the encoder and decoder. Consequently, the network produces much more accurate segmentation masks, either binary or multi-class, and thus can reveal the input image accurately on a pixel-by-pixel basis. It can generate segmentation masks (binary or multi-class) and as such, it is also able to provide fine-grained pixel-wise classification of input images. Figure 5.1 shows the U-Net architecture used.

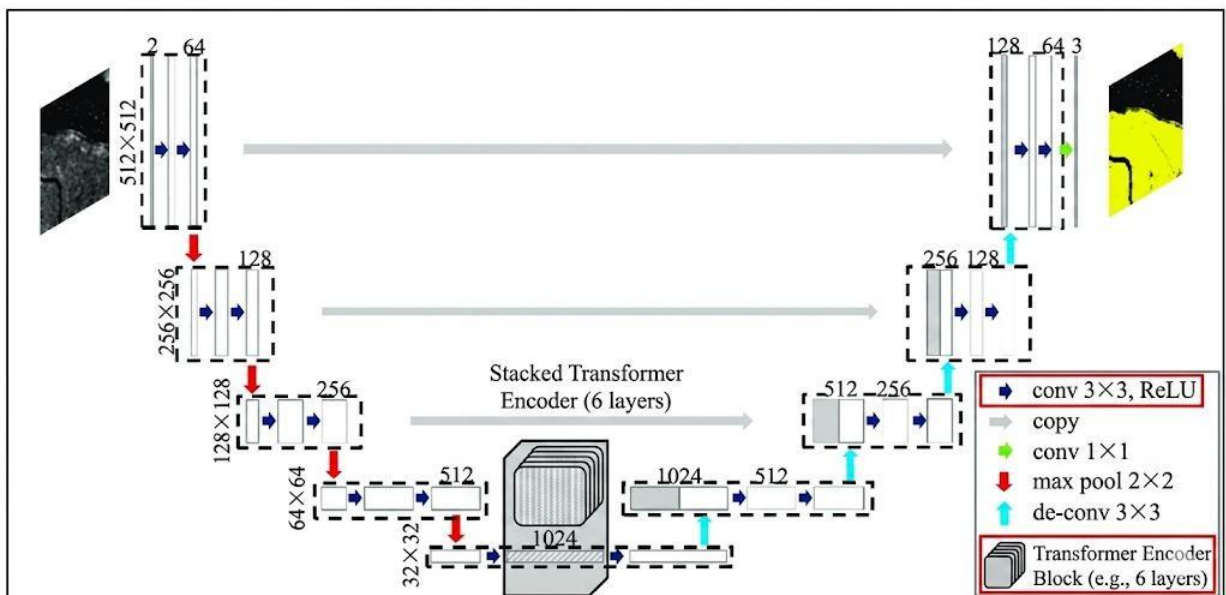


Figure 5.1: U-Net Architecture

5.1 DATA PREPARATION AND PREPROCESSING

The process starts with the ingestion and preparation of the raw medical imaging data for deep learning.

Ultrasound Image Dataset (800 x 540) & Data Annotation: The system uses raw ultrasound images (originally 800 x 540 resolution) and their corresponding ground truth masks which have been expertly labeled. The masks indicate the precise pixel coordinates of the fetal head.

Data Preprocessing: To speed up processing and ensure consistency, the images are resized to a square shape (512 x 512). The pixel intensities are also normalized to speed-up model training and make it less sensitive to differences in contrast of the ultrasound images.

Data Partitioning: The normalized data is split into Training, Validation and Test sets. This allows the model to be trained on one set, validated on another, and finally tested on a completely independent set to assess its learning capability.

5.2 DEVELOPMENT OF DEEP LEARNING MODEL

The System uses a U-Net model which is the standard for biomedical image segmentation. The model consists of three components: a contracting path (Encoder) to learn the visual context and enable feature map extraction, a bottleneck, and an expanding path (Decoder) which performs spatial localization to return the head.

Model weights are optimised using a combination of Binary Cross-Entropy (BCE) and Dice Loss. BCE calculates how correct the model is on pixel-wise classification and Dice loss tries to maximise the overlap between predicted mask and ground truth as, in our case due it has imbalanced classes i.e. background and head. The primary validation metric is Dice Coefficient.

After training the model is capable of predicting new images' segmentation mask, which will produce a binary mask where we have pixels as “head→1” or “background→0”.

5.3 MORPHOLOGICAL PROCESSING & GEOMETRIC EXTRACTION

However, the neural network predictions may at times have some minor artifacts; therefore, the process involves using mathematical morphology and geometry in order to extract valuable data from them.

Mask Processing: Firstly, the predictions are used in order to create a better binary mask (where only few pixels having low predictions are removed). Secondly, a fill algorithm will be applied in order to refine the final segment fetal head mask into solid objects.

Ellipse Generation from Mask Contours: Firstly, contour extraction allows the system to get

the outline of the mask through which it generates a contour of the head. Secondly, the generated contour is fitted to an ellipse as the fetus skull takes on an almost perfect shape of an ellipse. Thirdly, this coarse biological shape offers the system fixed geometrical axes for analysis.

Calculation of Parameters: An ellipse generates the digital measurement (Semi-Major Axis a & Semi-Minor Axis b). These are used to calculate the Head Circumference (HC), Biparietal Diameter (BPD), Occipitofrontal Diameter (OFD) and the Head Area. Importantly, the metadata CSV is utilized to input the actual distance between the pixels.

5.4 CLINICAL INFERENCE

Clinical inferences can then be made using the derived physical measurements.

Fetal Gestational Age Estimation: Comparison is done between the physical dimensions obtained from the measurement (HC, BPD, OFD) and obstetric growth curves/standardization regression models to estimate fetal gestational age (GA).

Growth Evaluation: This involves the classification of measurements with respect to their corresponding expected percentiles of GA. Outliers beyond the established confidence levels represent sheared morphological abnormalities.

5.5 ASSESSMENT & OUTPUT

The system's performance is evaluated and the results are collated in the final phase.

Statistical Ground Truth & Error Metrics: The automated Head Circumference (HC) is statistically compared to the clinical ground truth measurements found in the CSV that the dataset came from. The system reports error metrics like Mean Absolute Error (MAE) and percentage error to determine the system's clinical usefulness.

Final Output: The pipeline produces a clinical package for review, including a visual representation of the segmented mask, geometric measurements, the predicted Gestational Age (GA) and a diagnostic report (Normal/Abnormal) and a summary table of all results.

CHAPTER 6

RESNET50 ARCHITECTURE

ResNet50 is a deep convolutional neural network that comprises 50 layers with a method known as residual learning to address the issue of vanishing gradient in very deep networks. It does not explicitly learn a complete mapping between inputs and outputs, but rather learns the difference (residual) between them by enabling skip connections between the input of a layer and its output, which makes it easier to compute gradients during training. The architecture is designed based on bottleneck residual blocks with three convolutional layers (1×1 , 3×3 and 1×1) with the former layer decreasing the dimensions, the latter layer extracting, and the final layer restoring the dimensions. The network is structured into more stages with more depth and more feature channels with the first convolution and max pooling followed by stacked residual blocks and finally a global average pooling then a fully connected classification layer. This architecture allows ResNet50 to be trained on rich and complex image features, which has made it effective in tasks such as abnormality detection in medical imaging, where proper extraction of features in noisy ultrasound images is essential.

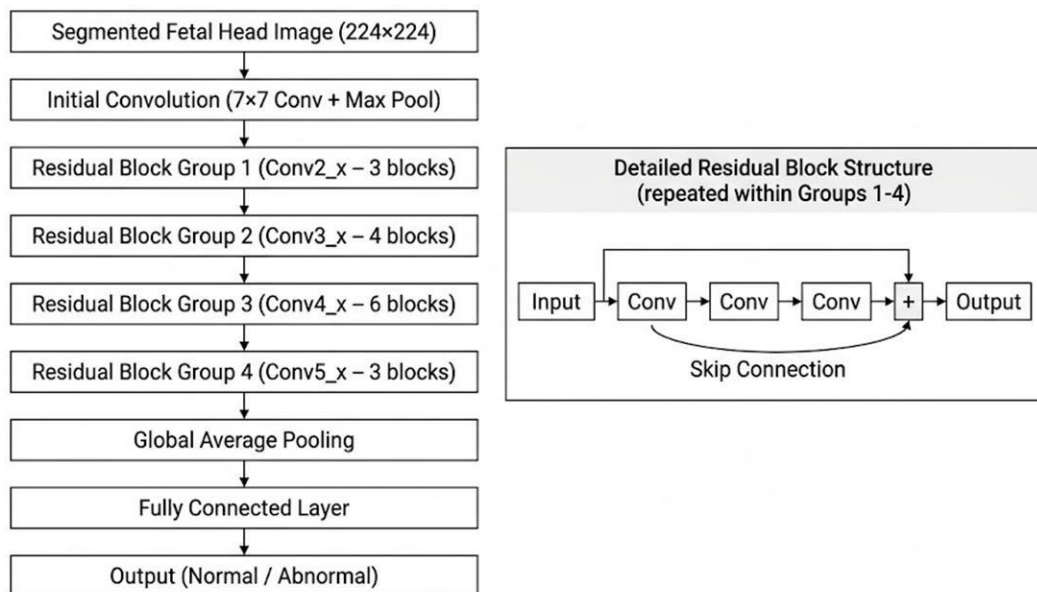


Figure 6.1: ResNet50 Architecture

6.1 DEEP LEARNING CLASSIFICATION ARCHITECTURE

This flow chart shows the structure of the convolutional neural network (CNN) applied to classify the images into Normal or Abnormal based on the features. This type of structure belongs to the ResNet-50 architecture, which has been highly modified for biomedical images. The architecture of the CNN consists of the following components: Initial Feature Extraction, Deep Residual Processing, and Classification.

6.2 INPUT AND INITIAL FEATURE EXTRACTION

Cropped & Segmented Fetal Head (224x224): The network receives the cropped & segmented fetal head image from the prior stage of the pipeline. The image is resized to 224x224 pixels, the typical size of the input to ResNet.

First Convolution (7x7 Conv + Max Pool): The image is first fed to a standard convolution block with a large 7x7 filter, and then a Max Pooling operation. This is a fast downsampler. It is a very aggressive downsampler of the image (e.g. width and height) and extracts low-level structural information (e.g. basic edges and textures).

6.3 DEEP RESIDUAL PROCESSING (THE CORE NETWORK)

The main part of the network is composed of four groups of residual blocks. As the network processes the data through these groups, it extracts higher- and higher-level features of the fetal skull.

Groups of Residual Blocks 1-4: These comprise 3, 4, 6 and 3 blocks respectively. As data moves from Group 1 to Group 4, the size of the feature maps continues to reduce but their "depth" (the number of channels of feature maps) grows.

6.4 RESIDUAL BLOCK STRUCTURE (INSET BOX)

The key element of this architecture is in the right inset box. In a conventional deep neural network, the more layers in the network, the more likely the "vanishing gradient" problem occurs, in which the network fails to learn.

The Skip Connection: To address this, the architecture incorporates a "Skip Connection" (or identity shortcut). This is where the input to the block is added (denoted by the + symbol) to the output of the convolutional layers before* it is passed to the next block.

Bottleneck Design: There are three Convolutional (Conv) layers within the block. This is referred to as a "bottleneck" architecture, which is usually designed with 1x1, 3x3 and 1x1 convolutions to reduce the dimensionality of the data, process it, and increase it back to its original size.

Why: The skip connection provides an avenue for information to flow directly to the output of

the model, allowing the gradients to flow more easily back through the network during training. This means that the network can be very deep (50 layers) without losing accuracy and can learn the very nuanced differences in the shape of normal and abnormal fetal heads.

6.5 THE CLASSIFICATION HEAD

When all the complex features are acquired by the remaining layers, interpretation of the acquired features is necessary to form the final output.

Global Average Pooling (GAP): Rather than converting the 3D feature map to a high dimensional 1D vector, which consumes a lot of computation power and will cause overfitting, GAP calculates the average of each feature map. In doing so, the parameters are reduced tremendously, yet retain the spatial information.

Fully Connected Layer: The fully connected layer acts as a regular classifier neural network that takes up the reduced features and extracts the necessary non-linear features.

Output (Normal/Abnormal): The final output layer uses an activation function to generate a value between 0 and 1 that determines whether the segmented fetal head is "Normal" or "Abnormal".

CHAPTER 7

RESULTS

The suggested Convolutional Neural Network with a U -Net architecture was trained to outline the fetal cranial contours in the obstetric ultrasound images. Table 7.1 shows the error metrics compared with the actual data.

Table 7.1: Error Metrics (Head Circumference)

Metric	Value
No. of test samples	5
Mean Absolute Error (MAE)	3.09mm
Mean Percentage Error (MAPE)	4.14%
Minimum Absolute Error	1.01 mm
Maximum Absolute Error	4.39 mm
Standard Deviation of Error	1.36 mm

After the segmentation, the process of contour extraction and ellipse fitting was followed to calculate the biometric parameters including Head Circumference (HC), Biparietal Diameter (BPD), Occipitofrontal Diameter (OFD) and Head Area. Table 7.2 shows us the model training performance and various loss function.

Table 7.2: Model Training Performance

Metric	Value
Final Training Loss	0.2303
Final Training Dice Coefficient	0.5923
Best Validation Loss	0.2892
Best Validation Dice Coefficient	0.4905

The estimated values of the means were in close agreement with the reference values hence justifying the dependability of proposed automated framework. The fact that fetal biometry can be controlled to measure rather accurately also supports the idea that deep learning-driven segmentation and the extraction of geometric parameters can be used to achieve accurate fetal biometry. Figure 7.1 displays the segmented output along with measurements of the fetus.

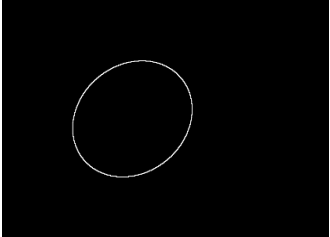
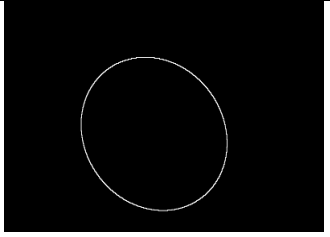
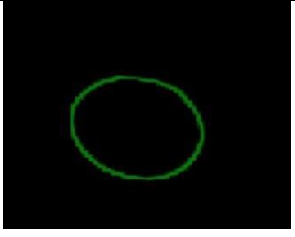
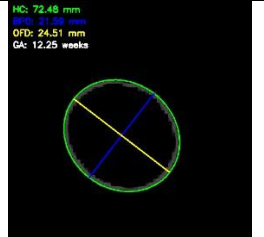
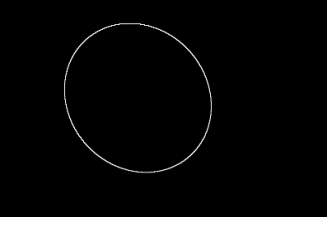
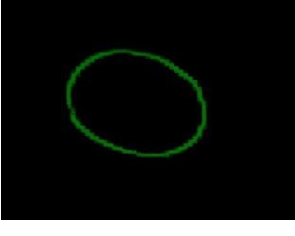
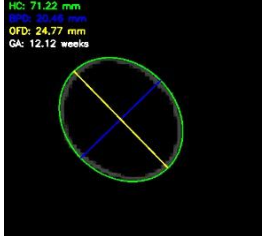
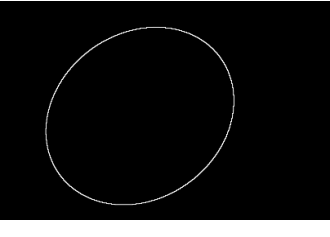
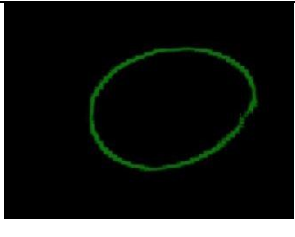
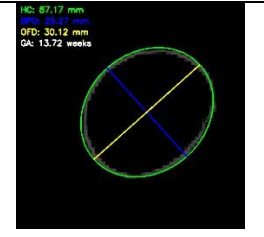
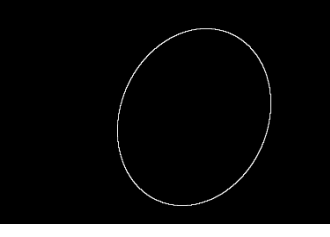
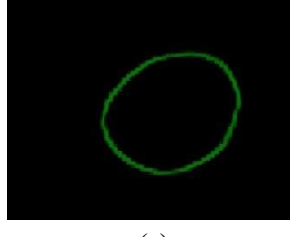
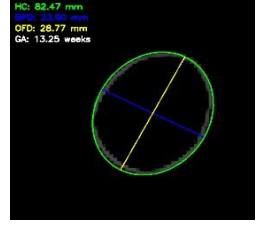
Input	Mask	Prediction	Measurement
			
			
			
			
			
(a)	(b)	(c)	(d)

Figure 7.1: Segmentation Result (a) Input image (b) Mask image (c) Model Prediction (d) Measurements

CHAPTER 8

CONCLUSION

This paper describes a deep-learning model that would automatically predict fetal cranial sizes based on the axial ultrasound images based on a segmentation methodology. The intersection-over-union (IoU) of the skull segmentations performed by the experts and labelled by the experts is improved by adding an initial layer of the U-Net-based segmentation followed by the feature refinement and geometric ellipse fitting steps, which leads to the abovementioned performance improvements. The procedure then matches major biometric indices of the head circumference (HC), biparietal diameter (BPD), occipitofrontal diameter (OFD), and the total cranial area in a standard manner. Unlike regression-based approaches which simply predict the target biometric measures, our framework first estimates anatomical boundaries. Subsequently, these boundaries are used in a geometric modeling phase to obtain the necessary measurements. The framework converts the dimensions obtained into millimetric dimensions by using a pixel-to-millimeter calibration. The method, therefore, provides clinically applicable measures.

The empirical data indicate that such segmentation approach is consistent with credible and meaningful biometric data, yielding small differences as compared to conventional clinical measures. The contour-based fitted ellipse also enhances the robustness of the system against ultrasound artefacts, including speckle noise and discontinuous edges. The framework automates the approach of quantifying cranial size of the foetus hence it have a clinical significance. This study intends to minimize the measurement errors associated with manual methods and increase the efficiency of measuring rodent models using geometric modeling based on segmentation using deep-learning ultrasound evaluations. Opportunities for future research include scaling of the system to other fetal measures, external validation on independent data, introduction of real-time applications and integration into clinical decision-support systems.

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Resources used at SRMIST

We would like to express our sincere gratitude to the Department of ECE, SRMIST, Kattankulathur for providing access to the _____ tools of Lab for _____ hours which were essential for the successful completion of this research/project.

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